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PERIPHERAL–CENTRAL EFFICACY DECOUPLING OF INCRETIN-BASED THERAPIES: A MECHANISTIC NARRATIVE REVIEW ACROSS THE CARDIO-HEPATIC AXIS AND THE NEURODEGENERATIVE BRAIN (2016–2026)

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1. ABSTRACT

Background: Glucagon-like peptide-1 receptor agonists (GLP-1 RAs) and dual GIP/GLP-1 co-agonists deliver rapid, reproducible clinical outcomes across the cardiovascular, renal, and hepatic compartments. Yet, after two decades of intense investigation, no incretin-based therapy has successfully demonstrated disease-modifying clinical efficacy in a pivotal phase 3 neurodegenerative disorder trial.

Objective: To dissect the specific micro-mechanistic and molecular reasons why robust peripheral incretin efficacy does not scale to the central nervous system (CNS), systematically separating drug-delivery constraints, receptor-architecture constraints, and cell-autonomous constraints.

Methods: We conducted a targeted PubMed/MEDLINE database search covering the last decade (1 January 2016 – 6 November 2026) utilizing the precise Boolean string: ("GLP-1" OR "semaglutide" OR "tirzepatide" OR "liraglutide" OR "exenatide" OR "lixisenatide") AND ("cardiovascular" OR "MACE" OR "NASH" OR "MASH" OR "fatty liver" OR "Alzheimer" OR "dementia" OR "Parkinson" OR "neuroprotection"). Eligible sources were human clinical trials, global registries, multi-disciplinary society consensus guidelines (AASLD, EASL–EASD–EASO), and primary preclinical studies mandatory to interpret macrovascular trial outcomes. This study represents a narrative mechanistic synthesis; no pooled statistics or meta-analytic computations were performed.

Results: Three non-redundant operational barriers explain the decoupling phenomenon. (1) Delivery barrier: Semaglutide does not measurably cross the intact endothelial blood–brain barrier (BBB), reaching homeostatic centers only via the fenestrated capillaries of circumventricular organs (CVOs) and tanycyte-mediated transport. Liraglutide penetrates partially via receptor-mediated endothelial transcytosis. Furthermore, oral platforms introduce a severe gastric bioavailability bottleneck, failing to establish the step-gradient required for cortical saturation. Thus, regional parenchymal exposure is low-to-negligible for the very agents used in the pivotal dementia trials. (2) Receptor-architecture barrier: CNS GLP-1R is expressed in hypothalamic nuclei (ARC, PVN, DMH), brainstem structures (NTS, area postrema), and scattered limbic structures, but is virtually absent from neocortical microcircuits and the BBB endothelium itself, ensuring that cognitive target fields are pharmacologically insulated. (3) Cell-autonomous barrier: Senescent neurons present pre-installed endosomal–lysosomal collapse, defective autophagic flux, and mitochondrial bioenergetic failure. Incretin signaling that robustly upregulates AMPK/mTOR/SREBP1 pathways to clear fat in mitotic liver environments cannot reboot the broken proteostatic clearance machinery of a non-dividing post-mitotic neural cell. Consistent with this triad, tirzepatide demonstrated profound histological MASH resolution and fibrosis regression in the SYNERGY-NASH trial (NEJM, 2024), whereas oral semaglutide failed both co-primary endpoints in the major phase 3 EVOKE/EVOKE+ Alzheimer trials (>3,800 participants) despite achieving peripheral target engagement.

Conclusion: We propose the term Peripheral–Central Efficacy Decoupling (PCED) to describe this systematic dissociation. We argue it represents a delivery–architecture–proteostasis problem rather than a class failure. We outline four testable predictions to falsify or refine this framework

and discuss structural bioengineering directions (including BBB-shuttled analogues and disease-stage stratification parameters) derived directly from it.

Keywords: GLP-1 receptor agonist; tirzepatide; semaglutide; MASH; SYNERGY-NASH; EVOKE; Alzheimer's disease; Parkinson's disease; blood–brain barrier; autophagy; Peripheral–Central Efficacy Decoupling.

2. INTRODUCTION

The incretin era has delivered one of the most comprehensive peripheral efficacy profiles in modern evidence-based medicine. Long-acting glucagon-like peptide-1 receptor agonists (GLP-1 RAs) reduce major adverse cardiovascular events (MACE) in type 2 diabetes and chronic obesity populations, slow chronic kidney disease progression, and reverse biopsy-proven metabolic dysfunction-associated steatohepatitis (MASH). The unimolecular dual GIP/GLP-1 co-agonist tirzepatide has compressed these multi-systemic systemic benefits into a single therapeutic sequence, prompting the American Association for the Study of Liver Diseases (AASLD) practice guidance to name GLP-1 RAs as a preferred pharmacological class for complex metabolic liver profiles.

Against this backdrop, the central nervous system (CNS) results read as a systematic, highly reproducible exception. Early-phase investigative signals were highly encouraging: liraglutide improved cerebral glucose metabolic parameters in a modest Alzheimer pilot (ELAD), and early mimetics like exenatide and lixisenatide displayed a limited capacity to slow motor decline in small Parkinson cohorts, though limited by half-life and tissue saturation barriers. Yet the clinical programme of evidence terminated in the largest test conducted to date: the phase 3 EVOKE/ EVOKE+ trials of oral semaglutide in early symptomatic Alzheimer's disease did not meet either co-primary cognitive endpoint, despite enrolling more than 3,800 global participants and achieving verified peripheral target engagement.

We argue this pattern is not noise but a clear biological signal that is mechanistically tractable. We further argue that the correct scientific frame is not "incretins fail in the brain" but "incretin efficacy is compartmentalized." The identical receptor class solves intense delivery, architectural, and proteostatic problems inside liver sinusoidal and vessel wall structures that it cannot solve within a senescent neocortex. To capture this clinical dissociation, we propose the term Peripheral–Central Efficacy Decoupling (PCED), defined as the reproducible dissociation between robust incretin target engagement and clinical efficacy in peripheral tissues versus the failure of the same engagement to produce measurable disease modification in CNS tissue, attributable to compounding delivery, receptor-architecture, and cell-autonomous constraints.

3. METHODS

Search architecture: The NCBI PubMed/MEDLINE repository was queried on November 6, 2026, with the active Boolean string: ("GLP-1" OR "semaglutide" OR "tirzepatide" OR "liraglutide" OR "exenatide" OR "lixisenatide") AND ("cardiovascular" OR "MACE" OR "NASH" OR "MASH" OR "fatty liver" OR "Alzheimer" OR "dementia" OR "Parkinson" OR "neuroprotection"), restricted to peer-reviewed documents published between 2016 and 2026 available in the English language. Supplementary verification utilized international registries (ClinicalTrials.gov), society guidelines (AASLD 2023 MASH guidance), and consensus nomenclature statements.

Selection logic: We deliberately adopted a rigid two-tier evidence architecture. Tier 1 (clinical anchors): Pivotal randomized controlled trials and large-scale longitudinal registries (e.g., SYNERGY-NASH 2024; EVOKE/EVOKE+ 2025; LEADER/SUSTAIN-6/SELECT-class cardiovascular outcomes trials). Tier 2 (mechanistic explanations): Preclinical, in vitro, and translational molecular models invoked solely to explain Tier-1 clinical outcomes, never conflated with them. This absolute separation is methodological: no molecular cascade reported from cell models is directly attributed to human trial data.

Synthesis: Narrative mechanistic synthesis. Where the underlying scientific literature was quantitatively homogeneous (such as the SYNERGY-NASH histology cohorts), raw baseline proportions are reported descriptively. No meta-analytic pooling or statistical interpolations were performed; this article is thus classified strictly as a narrative mechanistic synthesis.

Reporting integrity: Consistent with a strict integrity standard, every mechanistic assertion below is explicitly labeled as clinical-anchor evidence or mechanistic-support evidence. Two claims from early open-access drafts (radiolabel-confirmed deep semaglutide BBB penetration; GLP-1RA-driven direct closure of the hepatic CCL2/CCR2 axis via endothelial junctions) were downgraded or removed after direct verification because primary support could not be located in source registries. This transparent tracking forms an essential element of the review's core scientific argument.

4. RESULTS AND MECHANISTIC DEEP ANALYSIS

4.1 The peripheral axis: why liver and vessel wall respond

Clinical anchor. In the multi-center SYNERGY-NASH trial (phase 2, NEJM 2024), tirzepatide configurations achieved definitive MASH resolution without fibrosis worsening in 44%, 56%, and 62% of participants across 5, 10, and 15 mg cohorts, respectively, compared to just 10% in

the placebo arm. Furthermore, structural improvement of liver fibrosis by at least one stage without MASH worsening occurred in 51–55% of treated individuals versus 30% on placebo. Macrovascular and cardiovascular protectivity is anchored by the global GLP-1 RA CVOT clinical trials (LEADER, SUSTAIN-6, REWIND, SELECT), which document significant MACE reductions completely independent of absolute glucose lowering.

Mechanistic support. Hepatic steatosis reversal and subsequent endothelial recovery under incretin agonism are linked to the structural re-engagement of the intrahepatic AMPK/mTOR/SREBP1 metabolic axis. In metabolic humanized models, liraglutide and dual agonists drive hepatic AMPK phosphorylation, suppress p70S6K signaling, block SREBP1 nuclear translocation, and reduce de novo lipogenesis independently of localized weight loss. SREBP1 suppression simultaneously defuses the lipotoxic endoplasmic reticulum (ER) stress cascade within the organ. Concurrently, incretin target engagement on liver sinusoidal endothelial cells (LSECs) downregulates matrix metalloproteinase synthesis and pro-inflammatory transcripts via downstream ERK1/2/NF-κB pathways. Because hepatocytes and LSECs represent a highly responsive, mitotically active cell population with profound regeneration potential, the cardio-hepatic axis acts as a highly dynamic metabolic sink; removing peripheral lipotoxic stress allows immediate, linear structural tissue recovery.

4.2 The central axis: the delivery, architecture, and proteostasis triad

Delivery constraint and Bioavailability Barrier (Mechanistic support). Contemporary acylated incretin mimetics hit a severe pharmacokinetic bottleneck at the blood–brain barrier (BBB). Tracing protocols confirm that long-acting acylated peptides like semaglutide do not freely cross an intact endothelial BBB. Instead, central nervous system exposure is structurally restricted to the fenestrated capillaries of the circumventricular organs (CVOs) and slow, saturable tanycyte-driven transport in the median eminence. While early-generation mimetics like exenatide and lixisenatide displayed a limited capacity to modulate motor and cognitive signals in smaller pilots, their short systemic half-lives limited deep parenchymal tissue saturation. Conversely, liraglutide penetrates partially through receptor-mediated endothelial transcytosis.

Importantly, the translation to the oral semaglutide platform utilized in the pivotal EVOKE/EVOKE+ dementia trials introduces a stark oral bioavailability barrier. Formulating the peptide with salcaprozate sodium (SNAC) successfully drives gastric absorption to achieve peripheral target engagement, but it does not modify the molecule's intrinsic neuro-vascular permeability mechanics. At clinically achievable human oral doses, circulating plasma concentrations fail to generate the steep concentration gradient required to overcome the BBB and drive meaningful parenchymal or deep cortical saturation, leaving cognitive structures pharmacologically underserved.

Architectural insulation (Mechanistic support). Mapping the topological distribution of the human central GLP-1 receptor (GLP-1R) network reveals a severe mismatch with Alzheimer-relevant topography. GLP-1R density is highly clustered within homeostatic autonomic hubs: the arcuate nucleus (ARC), paraventricular nucleus (PVN), and dorsomedial hypothalamus (DMH), alongside the brainstem nucleus tractus solitarius (NTS) and area postrema. Neocortical microcircuits, hippocampal CA1 fields, and the microvascular BBB endothelium itself lack meaningful receptor populations. Thus, even if central drug delivery is maximized via alternative routes, the primary structural target fields for dementia intervention remain pharmacologically insulated from direct agonism.

Cell-autonomous collapse and Post-Mitotic Resistance (Mechanistic support). In peripheral compartments like hepatocytes, active incretin signaling reboots metabolic homeostasis because these cells possess a highly responsive lipophagy setup and mitotic replication potential. Conversely, senescent neurons present a permanent cell-autonomous collapse governed by their post-mitotic nature. Neurodegenerative states carry advanced endosomal–lysosomal enlargement, defective autophagic flux, and chronic mitochondrial bioenergetic failure. While incretin signaling can transiently upregulate AMPK and modulate microglial M1/M2 polarization profiles in adjacent glia, it cannot reboot the broken proteostatic clearance machinery of a non-dividing post-mitotic cell. Structural tau aggregation and amyloid-β fibrillation networks display massive therapeutic inertia because the structural damage inside post-mitotic neural architectures is irreversible via macrovascular metabolic offloading alone.

Table 1. Compartmentalized Micro-Environmental Differences: Periphery vs. CNS

Feature / Barrier	Peripheral Axis (Liver / Vessel Wall)	Central Axis (Neocortex / Cerebrum)
Primary Cell State	Mitotic / Highly Regenerative	Post-Mitotic / Irreversible Structural Damage
Drug Delivery Profile	Unhinderd Vascular Exposure	Restricted via BBB; Oral Bioavailability Barrier
Receptor Topology	Dense LSEC & Endothelial Network	Localized in CVOs/NTS; Absent in Neocortex
Autophagic Flux	Dynamic AMPK/mTOR Lipophagy	Chronic Lysosomal-Endosomal Collapse
Clinical Trial Status	SYNERGY-NASH (2024): 62% MASH Resolution	EVOKE/EVOKE+ (2026): Failed Co-Primary Endpoints

5. TESTABLE PREDICTIONS TO REFINE THE PCED FRAMEWORK

To preserve the falsifiability of the Peripheral–Central Efficacy Decoupling (PCED) model, we project four definitive experimental outcomes:

Prediction of BBB-Shuttle Superiority: If a frontier GLP-1RA conjugated to a transferrin receptor (TfR) antibody shuttle achieves uniform parenchymal neocortical concentrations in human pilots, it will significantly modulate neuro-inflammatory biomarkers (e.g., CSF sTREM2, plasma GFAP) at a fraction of the standard dose, confirming that delivery, not class mechanics, is the primary gatekeeper.

Prediction of Early-Stage Divergence: Disease-stage stratification screens will show that incretin-based therapy achieves a statistically meaningful delay in cognitive decline only when initiated during pre-clinical or prodromal phases (mild cognitive impairment with intact endosomal function), completely losing efficacy once macro-lysosomal collapse is established on structural tau imaging.

Prediction of Regional Tau Autonomy: Patients receiving ultra-high-dose incretin regimens will exhibit marked peripheral weight loss and reductions in cardiovascular calcification, yet sequential tau-PET tracks will confirm unhindered spread through neocortical microcircuits, validating the receptor-architecture insulation hypothesis.

Prediction of Multi-Agonist Synergistic Failure: Unimolecular triple agonists (GLP-1/GIP/Glucagon) will deliver unprecedented clearance of hepatic steatosis (<5% fraction on MRI-PDFF) but will show a definitive lack of clinical disease modification in late-stage Lewy body or Alzheimer's trials unless co-formulated with a direct cell-autonomous proteostasis rebooter.

6. DECLARATIONS & REPORTING INTEGRITY

Funding: The authors state that no specific funding or institutional grants were received for this study.

Conflicts of Interest: The authors declare that they have no competing financial or personal relationships that could influence the work reported in this paper.

Ethical Approval: Not applicable. This manuscript is a mechanistic review based strictly on secondary peer-reviewed published literature and contains no human participant datasets or primary animal tissue models.

Data Availability: The bibliographic and phenotypic metadata analyzed in this review are completely transparent and available within the PubMed/MEDLINE repository using the search criteria outlined in the methods section.